

OFFICIAL

Administration of Anaesthesia for Electroconvulsive Therapy Guideline

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1. Purpose

The Electroconvulsive therapy (ECT) utilises electrical energy to generate a brief pulse wave in order to elicit a safe and effective treatment for severe psychiatric and depressive disorders.

The provision of ECT in WA is strictly regulated under sections 194 to 199 of the Mental Health Act 2014. (1)

During the period of 2023-2024, 607 completed courses of ECT were reported to The Chief Psychiatrist in Western Australia. There were 6329 individual ECT treatments. (2)

The objective of anaesthesia is to provide a rapid onset and offset of both unconsciousness and muscle relaxation for the duration of the electrical stimulus and subsequent seizure, while minimising the physiological and physical effects.

The ECT service in Armadale Health Service (AHS) is a consultant led service organised by the psychiatric department. ECT procedures are planned and booked by the treating psychiatric consultants.

The Chief Psychiatrist's Guidelines for the Administration of Electroconvulsive Therapy (ECT) in Western Australia (version 2.2 March 2025) provides a framework for best practice in ECT provision in WA. (3)

The purpose of this guideline is to support safe and effective administration of anaesthesia for ECT at Armadale Health Service. This guidance covers: Anaesthetic assessment, consent, fasting, administration of anaesthesia and management and reporting of adverse events associated with ECT and post anaesthesia care.

Separate guidance exists regarding the delivery of electroconvulsive therapy in AHS. (4)

2. Applicability

This guideline applies Armadale Health Service Anaesthesia and Medical.

3. Guidelines

General Considerations for ECT

Anaesthesia for ECT should be administered by a specialist anaesthetist, i.e. registered medical practitioner, with Fellowship of the Australian and New Zealand College of Anaesthetists (FANZCA) or with equivalent qualifications credentialed to provide anaesthesia for ECT by Armadale Health Service.

They must comply with the requirements of the Australian and New Zealand College of Anaesthetists (ANZCA) and Australian Health Practitioner Regulation Agency (AHPRA) guidelines for practice.

In AHS, ECT must be provided inside operating theatres which fully comply with the ANZCA practice guidelines:

- PS55 'Position statement on minimum facilities for safe administration of anaesthesia in operating suites and other anaesthetising locations 2021' (5)
- PS54(A) 'Statement on the Minimum Requirements for Anaesthetic Machines and Workstations for Clinical Practice' (6)

When an anaesthetist does not have the necessary equipment or if there is a lack of staff to safely and efficiently administer general anaesthesia for ECT, or when the patient's medical condition dictates, the practitioner should refer the patient to another health care facility which can provide optimal care.

Standard precautions of infection control must be adopted for all ECT procedures which comply with ANZCA PG28 Guideline "Infection prevention and control in anaesthesia" (2025).(7)

4. Anaesthetic assessment

4.1 Anaesthesia Consultation, Evaluation and Consent

An anaesthetic consultation, evaluation and review will be requested to the Duty Anaesthetist (DA) by calling the phone number (0418949581) at least two to three days before the planned date of the first ECT session by the consultant psychiatrist in charge of ECT. At the same time, the ECT co-coordinator is to inform the theatre co-ordinator for booking and allocation. During the maintenance ECT sessions, when there is a change in the patient's medical status, medication or other conditions, the anaesthetic team must be informed via the DA.

The DA who receives the referrals will:

1. Review the patient
2. Determine the indication for ECT and any specific requirements that pertain to the proposed ECT Therapy
3. Review the anaesthetic history, drug doses and management of previous ECT sessions
4. Identify risk factors that may increase the perioperative risk and suggest measures that would minimise the risk including obtaining opinions from other consultants and specialities.

These specialties may include, but are not limited to, cardiology, vascular surgery, neurosurgery, neurology, endocrinology, ophthalmology, respiratory, bariatric surgery. Please note that "Electroconvulsive Therapy Procedure" states that as part of the pre ECT evaluation undertaken by the ECT Prescriber, in cases where doubt exists about a consumer's suitability for ECT, such as recent myocardial infarction or major cardiac event, the opinion of a specialist physician will be sought.

Appropriate and necessary investigations may need to be requested after taking a history and performing a physical examination. When the risk is higher than the benefit, cancellation of the proposed procedure may need to be considered in the patient's best interest.

5. Document any changes or suggestions and
6. Provide information to the patient regarding anaesthesia and possible complications,
7. Obtain an informed consent,

8. Inform the anaesthetist assigned for ECT treatment as required.
9. The DA will inform the anaesthetic team via email of the ECT booking as required. After the initial consultation, a copy of the completed part of the anaesthetic form must be placed in the patient's notes.

Pre-Anaesthesia Consultations should comply with ANZCA PG07 "Guideline on pre-anaesthesia consultation and patient preparation" (2024). (8)

4.2 High risk Electroconvulsive Therapy patients

Specific patients may have comorbidities which increase the risks of ECT. (3)

The seizure causes transient increase in intravascular, intracranial, intraocular and intrabdominal pressure and catecholamine release. Patients with pre-existing conditions can develop serious complications.

It may be appropriate to consult a specialist doctor (e.g. cardiologist).

Some patients may be considered too high risk for ECT. (3)

4.2.1 Conditions where Electroconvulsive Therapy can be harmful include but are not limited to:

Unstable or severe cardiovascular disease and conditions

Recent myocardial infarction within 3 months

- Unstable angina
- Poorly compensated heart failure
- Poorly Controlled Hypertension
- Severe valvular heart disease including critical aortic stenosis
- Significant arrhythmias
- Aneurysm or vascular malformation that might be susceptible to rupture with increased blood pressure
- Increased intracranial pressure - brain tumours or other space occupying cerebral lesions
- Recent cerebral infarction
- Severe chronic obstructive pulmonary disease, asthma, or pneumonia
- American Society of Anaesthesiologist (ASA) classification level 4 and above
- Pheochromocytoma
- Hyperthyroidism
- Retinal detachment
- Acute narrow angle glaucoma
- Osteoporosis
- Poor dentition
- Skull defect, titanium skull plate
- Patients with deep brain stimulators
- Post surgery
- Deep vein thrombosis

- Cochlear implant patient (may cause tissue damage to the cochlea or permanent damage to the implant)
- Systemic infection

5. Consent

The Mental Health Act 2014 requires approval from The Mental Health Tribunal (MHT) before ECT can be provided to a patient on an involuntary treatment order, children age 14-17 years or a person classified as mentally impaired accused. The MHA 2014 prohibits ECT being given to children under 14 years of age. [Appendix 1](#)(2)

Where emergency ECT is required to be performed on an adult involuntary patient or a person who is mentally impaired accused, approval from the Chief Psychiatrist or a delegate must be obtained prior to the ECT being performed. (2)

Voluntary patients must provide informed consent to ECT before receiving ECT.

Voluntary patients must also provide informed consent to anaesthesia (for ECT) and have a valid completed anaesthetic consent form before receiving anaesthesia for ECT.

The anaesthetic team must follow ANZCA guidance PS26(A) "Informed consent". (9)

There is a collaborative approach to ECT. The anaesthetist should inform both patients and psychiatrist regarding any risk factors related to anaesthesia or ECT which they identify.

This could have the following impacts:

- Further specialist input, e.g. cardiology, may be required.(3)
- A plan may be put in place to minimise risk
- ECT may be delayed
- The psychiatrist may need to collect a new consent for ECT after informing patient of additional ECT risks
- The mental health tribunal may need to be involved in the case of an involuntary patient
- The anaesthetist may need to take a new anaesthetic consent for ECT after informing patient of additional anaesthetic risks
- The decision may be made by the psychiatrist to not offer ECT to the patient
- The patient may withdraw their consent to ECT
- It may be decided to do ECT at a tertiary hospital

In complex cases or where doubt exists regarding the legal basis or capacity of a voluntary patient to provide anaesthetic consent please contact The Head of Department or The Director of Clinical Services at AHS who can seek further advice if required.

During the evaluation:

The anaesthetic form should be completed documenting history, physical status, physical examination, and specific concerns that may impact the proposed treatment and/or affect patient outcome

Premedication should be charted if necessary.

The anaesthetic consent form needs to be completed and signed by the DA and the patient at the time of evaluation before the ECT procedure. If the patient is incapable of providing consent,

the approval of the Mental Health Tribunal should be sighted and documented in liaison with the treating psychiatrist.

6. Pre-operative investigations and laboratory testing

Current state guidance states that ECG and serum electrolytes are essential investigations and should be performed before first scheduled ECT session (10). This is also reflected in local guidance (4). This may need to be repeated closer to the date as required.

Selected patients at risk of intracranial haemorrhage or raised intracranial pressure may also receive neuroimaging. (3) Other investigations may also have be requested.(3)

There is shared responsibility between the anaesthetist administering anaesthesia for ECT and the psychiatrist administering ECT to confirm the patient has had the correct investigations pre-procedure.

ECT itself can cause a number of life-threatening complications including cardiac arrhythmias which are usually transient but may persist.

Hyperkalaemia and hypokalaemia are two of the many causes of cardiac arrhythmia. Suxamethonium can cause or exacerbate hyperkalaemia in all patients. Patients at increased risk of hyperkalaemia with administration of suxamethonium include those with history of: neurological disease including those affecting upper motor neurones or lower motor neurones, paraplegia/quadruplegia, major burns, critical illness, renal failure, certain drugs including potassium, congenital or acquired myopathies, and rhabdomyosarcoma. Alternative muscle relaxants and reversal agents exist. (11)

7. Fasting

7.1 Fasting for Electroconvulsive Therapy

Minimum duration of fasting should be:

- 6 hours after a light meal
- 2 hours after clear fluids

This is consistent with Appendix 1 of ANZCA guideline PG07 Guideline on pre-anaesthesia consultation and patient preparation 2024 (12) and state-based ECT guidance (3)

If necessary, patients should be maintained on a level of observation sufficient to ensure compliance.

7.2 Patients at increased risk of aspiration

Should any patient risk factors for aspiration be present, the anaesthetist who completes the anaesthetic assessment described in part 1 of the guideline may elect to give the patient a modified fasting plan that prolongs the duration of fasting. ANZCA have published guidelines for prolonged fasting for patients taking glucagon-like peptide-1 receptor agonists (GLP-1RAs) and dual glucagon-like peptide-1 /glucose-dependent insulinotropic peptide receptor co-agonists. (14)

Anaesthetists may also prescribe premedications to reduce the acidity of gastric contents or promote gastric emptying (sodium citrate, ranitidine, esomeprazole, omeprazole, metoclopramide) after agreement with the psychiatrist.

Airway management options include inserting a definitive airway for ECT to protect against gastric aspiration. Elective intubation of ECT patients is exceedingly rare. If the decision is made to intubate a patient, then this will require a higher dose of anaesthetic drugs to achieve optimal conditions for intubation and prevent a hypertensive response to laryngoscopy.

Patients taking glucagon-like peptide-1 receptor agonists (**GLP-1RAs**) and dual glucagon-like peptide-1 /glucose-dependent insulintrophic peptide receptor co-agonists (or other medication that increases risk of aspiration) should be identified as early as possible in advance of ECT and discussed with the Anaesthetist. Patients taking these medications should be informed that these medications can increase the risk of aspiration during ECT and be provided with the best up to date information and clinical guidance to inform their decision to continue their current medication, switch to a safer alternative if available, or stop it. Patients should be informed that current ANZCA guidance recommends against elective pre-procedural cessation of these drugs. (14) Appropriate specialist input may need to be sought in the event a patient exercises their autonomy and requests to stop taking a GLP-1RA or related drug, for example consulting a diabetes specialist if they have diabetes.

The best up to date information and clinical guidance should be used to provide patients with fasting advice. Current ANZCA guidance recommends preprocedural diet modification with 24-hour clear fluid diet, followed by 6-hour fasting pre-procedure for all patients receiving GLP-1 RA's. (14) ANZCA have also published patient information leaflets on fasting for patients taking these drugs. (15)

Discussions and decisions about medication management, fasting guidance and any planned interventions to mitigate risk should be communicated to the multidisciplinary team including the anaesthetist administering anaesthesia for ECT.

Patients with a gastric band may need to be discussed with a bariatric surgeon. (3)

Patients at risk for relative dehydration should have an intravenous commenced early in the pre-ECT period.

All patients with diabetes should have a baseline glucometer reading performed. Thereafter, if indicated, a dextrose containing infusion should be commenced according to AHS peri-operative guidelines.

Regular medication should be continued during a course of ECT with a sip of water the morning of treatment.

If medications need to be withheld or changed, it should be done in liaison with Psychiatric team.

8. Administration of anaesthesia for Electroconvulsive Therapy

8.1 Team time out

WA state guidance recommends the following team time outs.(3)

1. Pre-anaesthetic time out
2. Post-anaesthetic, pre-treatment time out

8.1.1 Pre-anaesthetic time out

As a crucial safety measure, a time out procedure must be conducted before the anaesthetic to check and ensure:

- Correct patient
- Correct consent (Anaesthetic and ECT)
- Correct electrode placement by psychiatrist
- Correct pulse width by psychiatrist
- Correct dose by psychiatrist
- Correct anaesthetic drugs and doses
- False teeth removed
- Bite block available and ready to insert

The Team Time Out must be conducted jointly and simultaneously by the ECT practitioner, the ECT nurse and the anaesthetist, who must not be carrying out any other tasks at the same time. All non-essential communication should be ceased during the Team Time Out.

Everyone in the team must agree that each item is correct before moving onto the next check.

Any anticipated procedural or clinical difficulties which may affect the plan should also be briefly discussed so that their management is agreed by all.

The Team Time Out must be documented.

8.1.2 Post-anaesthetic, pre-treatment time out

This is a very brief time out to ensure the prevention of unmodified ECT.

Anaesthetist to confirm:

1. Patient anaesthetised
2. Muscle relaxant given
3. Flexible mouthguard in place

The anaesthetist should verbally and clearly state “anaesthetic delivered, ECT can proceed”. The ECT operator should say “ECT will be given/delivered now”.

8.2 Specific considerations

8.2.1 Possible increased risk of awareness under anaesthesia

Publications report awareness under anaesthesia with ECT (16)(17). The 5th national audit project (NAP5) reported an incidence of awareness under anaesthesia with muscle relaxation of 1:8200.(18) One retrospective study at a single centre found an incidence of awareness in patients undergoing ECT as high as 4.7%.(19)

Propofol and thiopentone, anaesthetic drugs used for ECT anaesthesia, have anticonvulsant properties which are dose dependent. ECT involves stimulating a therapeutic seizure. Generally, the anaesthetist will give the patient enough anaesthetic drugs to achieve general anaesthesia and unresponsiveness for the duration of treatment but avoid significantly overdosing on drugs which have anticonvulsant properties and could prevent or impair the therapeutic seizure. There is evidence of a clinically relevant inverse relationship between anaesthesia depth and quality of seizures in patients undergoing electroconvulsive therapy. (20)(21)

One publication states that any increased incidence of awareness that may be observed for ECT could be partly explainable because: ‘the induction dosage of IV anaesthetic agents tends to be as low as possible here [ECT] to... enable an adequate ECT response’.(22)

There are many reasons why a patient may not have a therapeutic seizure, and they may need a titration of stimulation intensity. The anaesthetist should be mindful of the risk of awareness with ECT anaesthesia. Patients should also be informed of the risk of awareness under anaesthesia when providing consent for anaesthetic.

Strategies that could be useful to prevent awareness under anaesthesia are:

- Performing a jaw thrust after induction agent to check for response before administering muscle relaxant (18) in addition to confirming no response to verbal stimuli and eyelash reflex
- Reviewing the anaesthetic record from previous treatments including drug doses and any comments
- Use of adjuncts including opioids or ketamine
- Administering anaesthetic drugs incrementally
- Minimising time between the onset of general anaesthesia and delivery of the electrical stimulus utilising the 'mind the gap' principle from NAP5 (18)
- Aiming to give more than the minimum dose of anaesthetic induction drugs

8.3 Muscle relaxation

Muscle relaxants are used to minimise the risk of a skeletal injury during the seizure. In combination with flexible bite block it also reduces risk of injury to teeth and tongue. Complete paralysis is not necessary (3), and the dose should be tailored based on the dose used in the patient's previous ECT and their weight. Muscle relaxation facilitates bag-mask ventilation.

8.4 Administration of anaesthesia for Electroconvulsive Therapy

The selection of drugs and doses should be individualized to account for each patient's unique requirements. For each course of ECT the drug doses received by the patient on previous days are recorded in the treatment record available in the patient notes.

Potential drug interactions with antidepressants (monoamine oxidase inhibitors [MAOI], selective serotonergic and noradrenergic reuptake inhibitors, lithium) and benzodiazepines must always be considered. The psychiatrist may decide to stop or change the psychiatric drugs patients is on in advance of ECT. (23)

The protection of the teeth and oral structures require special attention. The electrical stimulus results in direct stimulation of the masseter, pterygoid, and temporalis muscles and can cause an abrupt clenching of the jaw, despite muscle relaxation.

A flexible bite-block will be used to distribute the force of the contracting jaw, to enable protection of the teeth and other oral structures. All patients, including edentulous patients, require a bite-block (3) to be inserted.

Partial dentures may remain in as a support to protect single or vulnerable teeth. Loose dentures should be removed.

A hard plastic oropharyngeal airway should not be used as a bite-block.

The feet should be clear of the end of the bed (or end of bed removed) and bed rails should be lowered so that patient's limbs cannot make forceful contact with any hard structures during an induced seizure.

Managing other adverse events:

- The anaesthetist and psychiatrist should be informed of all side effects following ECT.
- Immediate side effects such as headache, myalgia, nausea, and drowsiness, should respond to symptomatic or supportive therapy.
- Muscle pain (myalgia) due to succinylcholine fasciculations usually lessens over the first few treatments in an ECT course and is uncommon later.
- Appropriate use of a mouthguard and adequate muscle relaxation should minimise the risk of tongue or dental injuries due to temporalis muscle stimulation. However substantial dental problems should be reviewed by a dentist. (3)

8.5 Physiological changes during Electroconvulsive Therapy

The electrical stimulus causes vagal stimulation. During and immediately following the electrical stimulus, sinus bradycardia, asystole, and hypotension may occur due to this pronounced parasympathetic surge. When a seizure occurs, significant levels of catecholamine are released into the circulation. The increased sympathetic outflow and adrenal catecholamine release during the seizure means that significant tachycardia and hypertension may occur. (3)

The evaluation of heart rate reveals a triphasic change from parasympathetic to sympathetic to parasympathetic cardiac autonomic activity after ECT stimulus (24).

Cardiac arrhythmias are frequent with ECT but are usually self-limiting. These arrhythmias include bradycardias, tachycardias and asystole. Specifically, the following rhythms have been observed: second degree AV block, ventricular tachycardia, ventricular fibrillation. (25)

During the seizure, cerebral blood flow increases markedly, oxygen extraction increases, and glucose metabolism increases. Cerebral auto regulation may be impeded, resulting in increased intracerebral pressure, intraocular and intragastric pressure increases.

In view of the serious risks involved in ECT, patients require careful evaluation before the decision to proceed with ECT.

8.6 Monitoring during anaesthesia

Monitoring during ECT Anaesthesia will comply with ANZCA PG18 Guideline on monitoring during anaesthesia 2025 (26):

- Continuous ECG
- Pulse oximetry
- Capnography
- Intermittent non-invasive blood pressure measurement
- Quantitative neuromuscular function monitor (e.g. train of four) should be available if needed

8.7 Anaesthetic procedure

Before starting an ECT list:

- Perform Anaesthetic machine and equipment check, and
- Ensure emergency drugs and apparatus are present, available, and functional.

During anaesthesia there must be a suitably trained anaesthetic assistant (anaesthetic technician or anaesthetic nurse) present.

Ensure that trolley/bed is capable of Trendelenburg positioning. Review the patient's chart including prior ECT anaesthetic records.

Ensure that the patient has an understanding of the proposed anaesthetic.

Discuss the planned procedure with the attending psychiatrist, including:

- Unilateral or bilateral electrode placement
- Necessity to titrate stimulus intensity
- Necessity to utilise proconvulsant drugs
- Requirement for limb isolation to observe motor manifestations of seizure
- The need for relative hyperventilation
- Establish intravenous access via an indwelling cannula.

Ensure there are no hard physical obstructions near the patient's feet or arms: feet clear of end of bed and rails lowered.

Ensure monitors are attached and obtain a baseline recording of parameters.

Administer anaesthetic drugs, ensuring adequate pre-oxygenation, airway control, and placement of the flexible bite-block. Anaesthetist must flush all IV cannulas with saline after administration of anaesthetic drugs and immediately after muscle relaxant.

Administer intermittent positive pressure ventilation with 100% oxygen until the electrical stimulus is administered. Continue post-stimulus until spontaneous and regular breathing are resumed.

Ensure the patient's positioning is optimal to ensure his or her safety.

The ECT stimulus may be applied only after the patient is adequately anesthetised, hemodynamically stable, and muscle relaxation is optimized. Use the post anaesthetic pre-treatment time out.

The anaesthetist must flush all IV cannulas with saline after administration of anaesthetic drugs and immediately after muscle relaxant.

During and immediately after stimulus, special attention must be directed to oxygenation and ventilation.

For hemodynamic monitoring, the blood pressure cuff should be cycled every 1–2 minutes.

When the seizure has terminated and haemodynamic stability achieved, the anaesthetist may choose to reposition the patient in the lateral position or keep them supine.

Once the patient is stable, rousable, and maintaining spontaneous ventilation, he or she may be transferred to the recovery area. Oxygen should be administered by facemask during transit.

The course of the anaesthetic including drugs given, haemodynamic monitoring, and other events should be recorded in the designated area of the anaesthetic record.

8.8 Anaesthesia drugs

8.8.1 Induction agents

The ideal induction agent would be short acting and have no effect on the seizure threshold or seizure duration. It would minimise haemodynamic changes associated with ECT.

The anaesthetist strives to achieve complete amnesia/unconsciousness throughout the period of muscle relaxation and seizure.

Propofol

Propofol is frequently used for ECT. (27) Do not add lignocaine to Propofol. Pain can be experienced on injection, which can be reduced by injecting into a fast-running intravenous placed into a larger bore vessel.

Propofol causes a greater attenuation of the cardiovascular changes of ECT compared to thiopentone.

Anticonvulsant properties reduces seizure duration significantly at moderate to large doses. (28)

Sodium Thiopentone

This barbiturate increases the seizure threshold in a dose-dependent fashion.

Ketamine

Ketamine has been used for induction of anaesthesia for ECT.

Due to its intrinsic sympathomimetic activity haemodynamic changes may be increased with the seizure. Ketamine can cause dysphoria on waking.

Ketamine can be combined with propofol, this may reduce the incidence and severity of hypertension and dysphoria that can sometimes be observed when ketamine is used as a single agent.

Ketamine's use as an adjunct to propofol for ECT is well described. It's use in ECT as an adjunct to propofol has recently been investigated with multiple randomised control trials both in The UK and Canada. (29)(30) Both of these trials demonstrated that the use of ketamine as an adjunct to propofol for ECT is safe. In The UK trial the anaesthetists were not blinded. Patients in the ketamine group received a 16% lower dose of propofol compared with the saline group.

Unlike propofol and thiopentone, Ketamine is not an anticonvulsant. Theoretically the use of ketamine as an adjunct may improve seizure duration by reducing propofol dose, but this has not been proven in practice.

8.8.2 Muscle relaxants

After the patient has received adequate anaesthesia and is unresponsive, the muscle relaxant is administered.

Muscle relaxants are used to minimize the risk of a skeletal injury during seizure. (3)

Neuromuscular monitoring allows a more accurate estimation of paralysis than clinical estimation.

Complete paralysis is neither desirable nor necessary (3)

Succinylcholine

Succinylcholine is the current relaxant of choice in a dose of 0.5 – 1.0 mg/kg.

Optimal relaxation occurs once all fasciculation has stopped.

If a repeat dose is required, an anticholinergic agent should be given before the succinylcholine, due to the risk of bradycardia.

Contraindicated in conditions with recent neurological deficit, malignant hyperthermia, hyperkalaemia or patient's at risk of hyperkalaemia (see risk factors above), burns, atypical pseudo cholinesterase, cholinesterase inhibition or known allergy.

Non depolarising Muscle relaxant (NDMR)

Rocuronium or other suitable NDMR may be used in cases where Suxamethonium is contra-indicated.

Sugammadex may be used for neuromuscular blockade reversal if rocuronium is used.

8.8.3 Anticholinergic medications

Atropine in a dose of 0.2 – 0.6 mg IV or Glycopyrrolate 0.2 – 0.4 mg IV may be used to decrease the bradycardia associated with the stimulus. They are recommended during the first treatment where the incidence of sub convulsive stimuli is higher while the convulsive threshold is evaluated. They should be administered intravenously in sufficient time (1 - 3 minutes) before the stimulus to attenuate the vagal effects on the heart.

Glycopyrrolate does not cross the blood–brain barrier, which reduces unwanted cognitive effects after ECT.(31).

8.8.4 Opioids

May reduce haemodynamic changes associated with ECT. Alfentanil and remifentanil both reduce the dose of propofol needed for anaesthesia for ECT and both increase seizure duration ($10\mu\text{gkg}^{-1}$ alfentanil or $1\mu\text{gkg}^{-1}$ remifentanil). They are not known to have anticonvulsant activity in therapeutic doses. (32,33)

8.9 Drug interactions and Electroconvulsive Therapy

Monoamine oxidase inhibitors may react with indirect acting sympathomimetic such as ephedrine and can cause a hypertensive crisis.

Cardiovascular complications including arrhythmia and hypo-/hypertension may be observed more frequently in patients taking certain medications. These include antidepressants, antipsychotics, anticholinesterase inhibitors, and cardiovascular drugs including antiarrhythmic and antihypertensive medications. All patients should have an ECG (3).

Selective serotonin reuptake inhibitors inhibit cytochrome isozymes and can therefore delay the clearance of antiarrhythmics and beta blockers.

Prolonged neuromuscular block with both depolarising and non-depolarising neuromuscular blockade can be seen in patients receiving Lithium.

Routine use of beta blockers for treating hypertension during ECT should be avoided as they can reduce seizure duration and have direct negative inotropic effect.

Esmolol (1mg/kg) and labetalol (0.3 mg/ kg) can be used to attenuate the sympathetic response of surge in systolic pressure and heart rate. Esmolol has a lesser effect on seizure duration than labetalol. (34)

Avoid using Verapamil as it has direct negative inotropic effect and it reduces atrio-ventricular conduction speed.

Dexmedetomidine 1 $\mu\text{g/kg}$ reduces haemodynamic responses with ECT but also reduces seizure duration. (35) Dexmedetomidine reduces myalgia associated with suxamethonium in patients undergoing ECT. (36)

8.10 Special cases

8.10.1 Pregnancy

ECT to pregnant patients is comparable in technique and administration pattern to non-pregnant.

ECT technique for pregnant patients should be modified as follows:

- Avoid hyperventilating the patient (hyperventilation diminishes foetal oxygenation by decreasing placental blood flow and reducing the dissociation of oxygen from haemoglobin)
- Hydrate patients with intravenous fluids prior to each ECT treatment
- For pregnancies between 12 to 23 weeks gestation, document the foetal heart rate before and after each ECT treatment
- If over 20 weeks place a wedge or lateral tilt to optimize maternal venous return, cardiac output, blood pressure, and uterine blood flow. Document the foetal heart rate before and after each ECT treatment
- If at least 24 weeks (foetus is considered viable if born) continuously monitor the foetal heart rate and uterine activity for 30 to 60 minutes before and after each ECT treatment.
- May require intubation to facilitate ventilation in a gravid patient and prevent aspiration (there is an elevated risk of aspiration associated with pregnancy). Extra care must be taken to avoid airway trauma which can lead to profuse bleeding secondary to oedema and hyper vascularity.

ECT may induce premature labour (oxytocin levels are elevated post ECT). Administer ECT only if resources for treating obstetric and neonatal emergencies are present.

8.10.2 Pacemakers and Automatic Implantable Cardioverter Defibrillators (AICD)

A consultant cardiologist should be consulted in patients who have a pacemaker or AICD where course of ECT is planned

The cardiology team / EP technician should deactivate AICDs before induction and reactivate the device after the seizure is complete.

The cardiologist may recommend changing pacemaker settings for duration of ECT.

The patient should have continuous ECG monitoring while the defibrillator is deactivated.

External defibrillator pads should be placed and a cardiac defibrillator available in case the patient develops an unstable rhythm

Magnets should be used with caution as they do not have the same effect on every pacemaker.

8.11 Documentation in anaesthetic record

Ensure that the doses of all drugs are recorded and any recommendations for change in doses at the next treatment are noted. Note comments regarding any complications or suggestions.

9. Management and reporting of adverse events associated with Electroconvulsive Therapy

9.1 Adverse events

Diagnose and treat abnormalities in vital signs and other complications, including, but not limited to

- Prolonged seizure
- Headache and Post-ictal delirium
- Nausea and vomiting
- Bronchospasm
- Arrhythmia, Angina, Hypo or Hypertension
- Hypo/hyperglycaemia in diabetic patients

A systematic review and meta-analysis published in Anaesthesiology in 2019 concluded that major adverse cardiac events after ECT are infrequent and occur in about one in 50 patients and after about one in 200 – 500 ECT treatments. The most commonly reported major adverse cardiac events were acute heart failure, arrhythmia, and acute pulmonary edema. All-cause mortality was 0.42 deaths per 1,000 patients and 0.06 deaths per 1,000 ECT treatments. Cardiac death accounted for 29% (23/79) of deaths. (37)

9.2 Complication specific to Electroconvulsive Therapy

9.2.1 Prolonged seizure

It is a potentially serious complication if the electrical seizure last for more than 120 seconds. It needs prompt recognition and treatment.

In case of prolonged seizure

Continue basic life support:

- Administer oxygen, continuously monitor and safely manage airway, breathing and circulation.
- Administer one-half the induction dose of the anaesthetic used if safe to do so.
- If this does not terminate seizure administer a benzodiazepine (eg midazolam 5mg)
- Call 55 and state “code blue medical emergency, and your location” to activate The Medical Emergency Team.
- Consider deepening the anaesthetic further using propofol, thiopentone and/or Volatile anaesthetic agents.
- If these medications do not terminate the seizure discuss with intensive care and psychiatrist and commence other anticonvulsant drugs (eg levetiracetam or phenytoin) if not contraindicated and follow instructions for safe administration

Ways that a prolonged seizure may be recognised could include:

- Witnessed patient movement or posturing
- EEG monitoring (from ECT machine)

- Recognition by the anaesthetist or team
- Patient failing to regain consciousness

Benzodiazepines such as midazolam are effective anticonvulsants.

Anaesthetic hypnotic drugs have anticonvulsant effects. This includes: propofol, thiopentone, and volatile anaesthetic agents. The anticonvulsant effect is dose dependant.

Anaesthetists are experts in provision of general anaesthesia and in managing airway, breathing and circulation safely.

Note that seizure related hypertension may make intubation unsafe due to risk of intracranial haemorrhage

The patient may need to be transferred to ICU following assessment by the intensive care team.

If there are concerns about reduced conscious level, failure to maintain airway or ongoing seizure or seizure risk, then the patient should be admitted to intensive care.

Note: muscle relaxants will stop the motor component of a seizure but not the seizure itself. In view of this, if a long-acting muscle relaxant is administered (if the patient is intubated) consider subclinical seizures (these may be apparent on the EEG) or consider reversing a long acting muscle relaxant to unmask seizure activity.

9.2.2 Head and post-ictal delirium

Inform the psychiatrist if a patient develops headache or post-ictal delirium. They can advise on treatment.

These patients should have a comprehensive assessment by an anaesthetic consultant including neurological examination at the time of first symptoms and further assessment to assess for any delayed progression of symptoms or signs.

Simple analgesics including paracetamol (if not contraindicated) are useful for headache and should lead to full resolution of symptoms at re-assessment. Consider admission to a medical bed with regular neurological observation performed by nurses and consider imaging such as CT head to detect rare complications such as intracranial haemorrhage if clinically indicated or if symptoms presenting after ECT do not improve or fully resolve.

9.3 Reporting

The Chief Psychiatrist WA Report 2023-2024 revealed that an adverse event during one or more treatments was reported for 12.6% of all ECT courses(2).

Adverse events that need to be reported to WA Chief Psychiatrist include:

- Premature consciousness during a treatment
- Anaesthetic complications
- Cardiac arrhythmia
- An acute and persistent confused state during recovery from a treatment
- Muscle tears or vertebral column damage
- Severe and persistent headaches
- Persistent memory deficit.

A previous Chief Psychiatrist WA Report (38) reported:

- Headache 60%
- Memory deficit 13 %
- Confused state 12%
- Anaesthetic complication 10%
- Premature consciousness/awareness 1%.

Note serious complications in the patient's hospital ECT record and communicate them to the patient's psychiatrist for reporting.

10. Post anaesthetic period

Post-Anaesthesia care should comply with the ANZCA Guideline PS04(A) "Position statement on the post-anaesthesia care unit (2020)".(39)

The anaesthetist should handover the patient to recovery staff in accordance with PS53(A) Position statement on the handover responsibilities of the anaesthetist.(40)

In accordance with the above guidance, the anaesthetist should remain in the vicinity until the patient is assessed as being stable and thus may be left in the care of post anaesthesia care unit (PACU) staff and PACU staff have explicitly accepted care of the patient. Handover or delegation of care in PACU should not proceed if airway/respiratory and cardiovascular status is not stable. They should be available for consultation during the recovery period.

Observations should be recorded at clinically indicated intervals and should include level of consciousness, oxygen saturation, respiratory rate, pulse rate and blood pressure. Blood glucose should be monitored if the patient has diabetes.

Interventions to prevent apnoea and related complications in PACU

- The anaesthetist must flush all IV cannulas after administration of anaesthetic drugs including muscle relaxant before handing patient over to recovery room staff.
- The anaesthetist to inform PACU staff that all cannulas are flushed in PACU handover (if this is the case)
- Intravenous access should be removed before leaving PACU.

The recovery nurse should raise any concerns and report any abnormalities to the anaesthetist or Duty Anaesthetist promptly, if required they can escalate by calling The Medical Emergency Team.

The patient should remain in PACU at least until they are alert and orientated, and their observations are within their own pre ECT range.

The patient must be free of complications and have their vital signs returned to acceptable limits before discharge to the Mental Health Unit. If the patient is admitted as a Day care case, the patient should also fulfil AHS same day surgical patient discharge criteria before discharge from the hospital.

11. Legislative context

The following documents are required to give effect to this policy:

Health Department of Western Australia (HDWA) Mandatory Policy

- [Western Australia Mental Health Act 2014](#)

- [WA Guardianship and Administration Act \(1990\)](#)
- [Health Practitioner Regulation National Law \(WA\) Act 2010](#)
- [Western Australia Carers Recognition Act 2004](#)
- [Clinical Incident Management Policy \(2015\)](#)
- [Clinical Alert Policy MP 0053/17](#)
- [Clinical Handover Policy MP 0095](#)
- [Consent to Treatment Policy](#)
- [Complaints Management Policy MP 0130/20.](#)

12. Supporting information

The following documents support the implementation of this policy:

- Armadale Kalamunda Group related policy / procedures
 - [Electroconvulsive Therapy Policy](#)
 - [Electroconvulsive Therapy Procedure](#)

13. Compliance, monitoring and evaluation

The Comprehensive Care Committee will be responsible for ensuring compliance against this guideline, evaluated, against the ECT monthly audits.

14. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 1 – Action 1.18.1
- Standard 1 – Action 1.83.1
- Standard 4 – Action 4.06.1

National Standards for Mental Health Services (2010):

- Standard 1: Rights and Responsibilities
- Standard 3: Consumer and Carer Participation
- Standard 6: Consumers
- Standard 8: Governance, leadership and management

[Chief Psychiatrist Standards for the Administration of Electroconvulsive Therapy \(ECT\) in Western Australia 2025](#)

- Standard 1: ECT Service delivery
- Standard 2: Consent and the legal framework
- Standard 3: Treatment provision
- Standard 4: Safety, quality and reporting

15. References

1. Mental Health Act 2014 (MHA14)
Legislation available from: [Mental Health Act 2014 - \[01-g0-02\].pdf](#)

2. Annual Report of the Chief Psychiatrist of Western Australia Chief Psychiatrist of Western Australia ENSURING SAFE AND HIGH-QUALITY MENTAL HEALTH CARE [Internet]. [cited 2025 Jul 8]. Available from: <https://www.chiefpsychiatrist.wa.gov.au/wp-content/uploads/2024/09/Chief-Psychiatrist-Annual-Report-2024.pdf>
3. Chief Psychiatrist's Guidelines for the use of Electroconvulsive Therapy in Western Australia 2024. Available from: <https://www.chiefpsychiatrist.wa.gov.au/wp-content/uploads/2025/03/CP-Guidelines-for-the-use-of-ECT-in-WA-ECT-Guidelines-V2.2-March-2025.pdf>
4. ECT Nurse Coordinator. Electroconvulsive Therapy Procedure [Internet]. HealthPoint-Policies@health.wa.gov.au. East Metropolitan Health Service, WA Health; 2025 [cited 2025 Jul 8]. Available from: <https://healthpoint.hdwa.health.wa.gov.au/policies/Policies/EMHS/AHS/MHSP.ECTProcedure.pdf>
5. PS55 Position statement on minimum facilities for safe administration of anaesthesia in operating suites and other anaesthetising locations 2021 Short title: Minimum safe facilities [Internet]. Available from: [https://www.anzca.edu.au/getContentAsset/9b0c33f8-6a0f-4f76-9e90-a751dc181f12/80feb437-d24d-46b8-a858-4a2a28b9b970/PS55-Minimum-safe-facilities-2021-\(Apdx-3-2024\).pdf?language=en](https://www.anzca.edu.au/getContentAsset/9b0c33f8-6a0f-4f76-9e90-a751dc181f12/80feb437-d24d-46b8-a858-4a2a28b9b970/PS55-Minimum-safe-facilities-2021-(Apdx-3-2024).pdf?language=en)
6. PS54(A) Position statement on the minimum safety requirements for anaesthesia machines and workstations for clinical practice 2021 Short title: Anaesthesia machine safety [Internet]. Available from: [https://www.anzca.edu.au/getContentAsset/b8d36123-1cb6-4180-a711-e170b815062b/80feb437-d24d-46b8-a858-4a2a28b9b970/PS54\(A\).PDF?language=en&view=1](https://www.anzca.edu.au/getContentAsset/b8d36123-1cb6-4180-a711-e170b815062b/80feb437-d24d-46b8-a858-4a2a28b9b970/PS54(A).PDF?language=en&view=1)
7. PG28 Guideline on infection prevention and control in anaesthesia 2025 Short title: Infection prevention and control [Internet]. [cited 2025 Jul 8]. Available from: <https://www.anzca.edu.au/getContentAsset/41b75be4-4c78-4318-8ac4-3d0dd2ca8f8f/80feb437-d24d-46b8-a858-4a2a28b9b970/PG28-Infection-prevention-and-control-2025-PILOT.pdf?language=en>
8. PG07 Pre-anaesthesia consultation [Internet]. Available from: <https://www.anzca.edu.au/getContentAsset/d4eb4cab-69e6-4228-8568-bbc911c6c505/80feb437-d24d-46b8-a858-4a2a28b9b970/PG07-Pre-anaesthesia-consultation-2024.pdf?language=en>
9. PS26(A) Position statement on informed consent for anaesthesia or sedation 2021 Short title: Informed consent [Internet]. Available from: [https://www.anzca.edu.au/getContentAsset/fde86aaf-c19c-4d71-8156-015d1d7b2c8d/80feb437-d24d-46b8-a858-4a2a28b9b970/PS26\(A\)-Position-statement-on-informed-consent-for-anaesthesia-or-sedation-2021.PDF?language=en&view=1](https://www.anzca.edu.au/getContentAsset/fde86aaf-c19c-4d71-8156-015d1d7b2c8d/80feb437-d24d-46b8-a858-4a2a28b9b970/PS26(A)-Position-statement-on-informed-consent-for-anaesthesia-or-sedation-2021.PDF?language=en&view=1)
10. Chief Psychiatrist's Standards for the Administration of Electroconvulsive Therapy (ECT) in WA [Internet]. Available from: <https://www.chiefpsychiatrist.wa.gov.au/wp-content/uploads/2025/01/ECT-Standards-2025.pdf>
11. Hovgaard HL, Juhl-Olsen P. Suxamethonium-Induced Hyperkalaemia: A Short Review of Causes and Recommendations for Clinical Applications. Edelstein S, editor. Critical Care Research and Practice. 2021 Feb 25;2021:1–6.
12. PG07 Guideline on pre-anaesthesia consultation and patient preparation 2024. Appendix 1- Fasting guideline [Internet]. Available from: <https://www.anzca.edu.au/getContentAsset/d4eb4cab-69e6-4228-8568-bbc911c6c505/80feb437-d24d-46b8-a858-4a2a28b9b970/PG07-Pre-anaesthesia-consultation-2024.pdf?language=en>
13. Office of The Chief Psychiatrist, South Australia. ECT Chief Psychiatrist (SA) Policy Guideline 2020 [Internet]. Available from: <https://s3-ap-southeast->

- 2.amazonaws.com/sahealth-ocp-assets/general-downloads/20200623-ECT-Policy-FINAL-edition-2-version-2-pre-PDF-version.pdf
14. ANZCA Clinical Practice Recommendations regarding patients taking GLP-1 receptor agonists and dual GLP-1/GIP receptor co-agonists prior to anaesthesia or sedation for surgical and endoscopic procedures. <https://www.anzca.edu.au/getContentAsset/0f35028e-e371-4220-a49a-ddee877051c8/80feb437-d24d-46b8-a858-4a2a28b9b970/Clinical-Practice-Recommendations-Periprocedural-GLP-1RA-use-Apr-2025.pdf?language=en>.
 15. ANZCA GLP-1 agonist patient information sheet [Internet]. Available from: https://www.anzca.edu.au/getContentAsset/04ccb059-2f86-4bb2-940b-a085501e5ae7/80feb437-d24d-46b8-a858-4a2a28b9b970/ANZCA_GLP-1_Agonist_info-sheet_V5.pdf?language=en
 16. Patient perspectives on intraoperative awareness with explicit recall: report from a North American anaesthesia awareness registry Kent, C.D. et al. *British Journal of Anaesthesia*, Volume 115, i114 - i121
 17. Huoponen SH, Sisa K, Saari T, Taittonen M, Ahlmén-Laiho U. The Recognition and Management of Adverse Effects in Electroconvulsive Therapy: Findings From a Finnish Survey Study. *Brain Behav.* 2025 Jan;15(1):e70227. doi: 10.1002/brb3.70227. PMID: 39829102; PMCID: PMC11743972.
 18. J. J. Pandit, J. Andrade, D. G. Bogod, J. M. Hitchman, W. R. Jonker, N. Lucas, J. H. Mackay, A. F. Nimmo, K. O'Connor, E. P. O'Sullivan, R. G. Paul, J. H. M. G. Palmer, F. Plaat, J. J. Radcliffe, M. R. J. Sury, H. E. Torevell, M. Wang, J. Hainsworth, T. M. Cook, James Armstrong, Jonathan Bird, Alison Eddy, William Harrop-Griffiths, Nicholas Love, Ravi Mahajan, Abhiram Mallick, Ian Barker, Anahita Kirkpatrick, Jayne Molodynski, Karthikeyen Poonnusamy, Ramani Moonesinghe, David Weatherill, David Smith, Maddy Humphrey, Mary Casserly, Sharon Drake, Jairaj Rangasami, on behalf of the Royal College of Anaesthetists and the Association of Anaesthetists of Great Britain and Ireland, 5th National Audit Project (NAP5) on accidental awareness during general anaesthesia: summary of main findings and risk factors, *BJA: British Journal of Anaesthesia*, Volume 113, Issue 4, October 2014, Pages 549–559, <https://doi.org/10.1093/bja/aeu313>
 19. Welch, Marnie Bowen MD*; Armella Tangarife, Daniela BA†; Knight, Julia C. MD, PhD‡. A Retrospective Study of Intraoperative Awareness in Patients Undergoing Electroconvulsive Therapy at Dartmouth Hitchcock Medical Center Over a 3-Year Period. *The Journal of ECT* ():10.1097/YCT.0000000000001143, April 28, 2025. | DOI: 10.1097/YCT.0000000000001143
 20. Gasteiger, Lukas MD*; Heil, Michel MD†; Hörner, Elisabeth MD*; Andexer, Janine*; Kemmler, Georg PhD†; Hausmann, Armand MD†; Lederer, Wolfgang MD*. Relationship Between Anesthesia Depth and Quality of Seizures in Patients Undergoing Electroconvulsive Therapy: A Prospective Observational Study. *The Journal of ECT* 38(1):p 62-67, March 2022. | DOI: 10.1097/YCT.0000000000000792
 21. Guerrier G, Gianni MA. The effectiveness of BIS monitoring during electro-convulsive therapy: A systematic review and meta-analysis. *J Clin Anesth.* 2019 Dec;58:100-104. doi: 10.1016/j.jclinane.2019.05.006. Epub 2019 May 28. PMID: 31151038.
 22. Acharya H, Gaur A, Kunigiri G. Anesthetic and psychiatric implications of accidental awareness under general anesthesia during electroconvulsive therapy. *Saudi J Anaesth.* 2018 Apr-Jun;12(2):335-338. doi: 10.4103/sja.SJA_675_17. PMID: 29628852; PMCID: PMC5875230.
 23. Naguib M, Koorn R. Interactions between psychotropics, anaesthetics and electroconvulsive therapy: implications for drug choice and patient management. *CNS Drugs.* 2002;16(4):229-47. doi: 10.2165/00023210-200216040-00003. PMID: 11945107.

24. Suzuki Y, Miyajima M, Ohta K, Yoshida N, Okumura M, Nakamura M, Sasano T, Kawara T, Matsuura M, Matsushima E. A Triphasic Change of Cardiac Autonomic Nervous System During Electroconvulsive Therapy. *J ECT*. 2015 Sep;31(3):186-91. doi: 10.1097/YCT.0000000000000227. PMID: 26307953.
25. Elias A, Das S, Kirkland J, Loyal S, Thomas N. Safety of electroconvulsive therapy in the context of physiological and medical complexity: A state-of-the art review. *PCN Rep*. 2025 Jan 3;4(1):e70051. doi: 10.1002/pcn5.70051. PMID: 39759458; PMCID: PMC11695668.
26. PG18 Guideline on monitoring during anaesthesia 2025 [Internet]. Available from: [PG18\(A\)-Guideline-on-monitoring-during-anaesthesia-2017.PDF](#)
27. Huoponen, Saara H. MD*; Sisa, Katrin MD*; Taittonen, Markku MD*; Saari, Tom MD*; Ahlmén-Laiho, Ulla D Med Sci, MD*,†. A Survey of ECT Anesthesia Practice Variation in Finland. *The Journal of ECT* 41(2):p 84-88, June 2025. | DOI: 10.1097/YCT.0000000000001042
28. Rasmussen KG. Propofol for ECT anesthesia a review of the literature. *J ECT*. 2014 Sep;30(3):210-5. doi: 10.1097/YCT.0000000000000093. PMID: 24820943.
29. Anderson IM, Blamire A, Branton T, Clark R, Downey D, Dunn G, Easton A, Elliott R, Elwell C, Hayden K, Holland F, Karim S, Loo C, Lowe J, Nair R, Oakley T, Prakash A, Sharma PK, Williams SR, McAllister-Williams RH; Ketamine-ECT Study team. Ketamine augmentation of electroconvulsive therapy to improve neuropsychological and clinical outcomes in depression (Ketamine-ECT): a multicentre, double-blind, randomised, parallel-group, superiority trial. *Lancet Psychiatry*. 2017 May;4(5):365-377. doi: 10.1016/S2215-0366(17)30077-9. Epub 2017 Mar 27. PMID: 28359862; PMCID: PMC5406618.
30. Woolsey AJ, Nanji JA, Moreau C, Sivapalan S, Bourque SL, Ceccherini-Nelli A, Gragasin FS. Low-dose ketamine does not improve the speed of recovery from depression in electroconvulsive therapy: a randomized controlled trial. *Braz J Psychiatry*. 2022 Jan-Feb;44(1):6-14. doi: 10.1590/1516-4446-2020-1705. PMID: 34076068; PMCID: PMC8827368.
31. Zolezzi M. Medication management during electroconvulsant therapy. *Neuropsychiatr Dis Treat*. 2016 Apr 19;12:931-9. doi: 10.2147/NDT.S100908. PMID: 27143894; PMCID: PMC4844444.
32. Nguyen TT, Chhibber AK, Lustik SJ, Kolano JW, Dillon PJ, Guttmacher LB. Effect of methohexitone and propofol with or without alfentanil on seizure duration and recovery in electroconvulsive therapy. *Br J Anaesth*. 1997 Dec;79(6):801-3. doi: 10.1093/bja/79.6.801. PMID: 9496217.
33. Akcaboy ZN, Akcaboy EY, Yigitbası B, Bayam G, Dikmen B, Gogus N, Dilbaz N. Effects of remifentanil and alfentanil on seizure duration, stimulus amplitudes and recovery parameters during ECT. *Acta Anaesthesiol Scand*. 2005 Sep;49(8):1068-71. doi: 10.1111/j.1399-6576.2005.00766.x. PMID: 16095445.
34. Beta-blocking agents during electroconvulsive therapy: a review. Boere, E. et al. *British Journal of Anaesthesia*, Volume 113, Issue 1, 43 – 51
35. Subsoontorn P, Lekprasert V, Waleeprakhon P, Ittasakul P, Laopuangsak A, Limpoon S. Premedication with dexmedetomidine for prevention of hyperdynamic response after electroconvulsive therapy: a cross-over, randomized controlled trial. *BMC Psychiatry*. 2021 Aug 17;21(1):408. doi: 10.1186/s12888-021-03406-9. PMID: 34404384; PMCID: PMC8371895.
36. Sriramka B, Panigrahy S, Ramasubbu MK, Mishra SN. Dexmedetomidine for reducing succinylcholine-induced myalgia in patients undergoing electroconvulsive therapy: A randomised controlled trial. *Indian J Anaesth*. 2024 Jun;68(6):560-565. doi: 10.4103/ija.ija_1159_23. Epub 2024 May 8. PMID: 38903251; PMCID: PMC11186527.

37. Duma A, Maleczek M, Panjikaran B, Herkner H, Karrison T, Nagele P. Major Adverse Cardiac Events and Mortality Associated with Electroconvulsive Therapy: A Systematic Review and Meta-analysis. *Anesthesiology*. 2019 Jan;130(1):83-91. doi: 10.1097/ALN.0000000000002488. PMID: 30557212; PMCID: PMC6300062.
38. Ensuring Safe and High Quality Mental Health Care' Annual Report of the Chief Psychiatrist of Western Australia 01 July 2023 – 30 June 2024 [Internet]. Available from: [Office of the Chief Psychiatrist – Annual Report 2023-2024](#)
39. PS04(A) Position statement on the post-anaesthesia care unit 2020 [Internet]. Available from: [PS04\(A\)-Position-statement-on-the-post-anaesthesia-care-unit-2020.PDF](#)
40. PS53(A) Position statement on the handover responsibilities of the anaesthetist [Internet]. Available from: [PS53\(A\)-Position-statement-on-the-handover-responsibilities-of-the-anaesthetist-2013.PDF](#)

16. Document owner

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17. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4	03 11 2025	03 11 2025	Full review, updates position statement on minimum facilities for safe administration of anaesthesia and appendix 1 (flowchart for consent process for anaesthesia and ECT) updated.

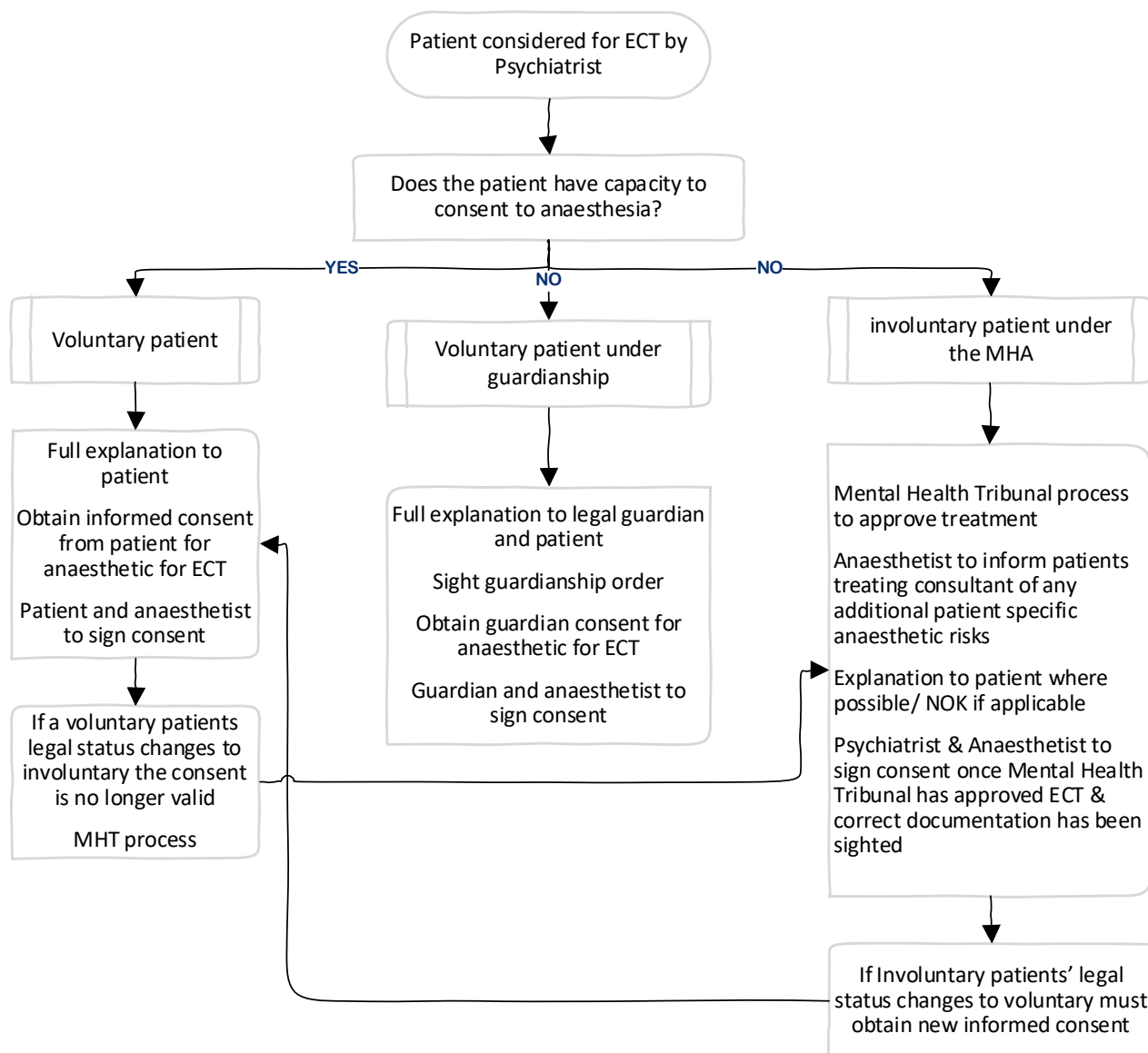
18. Authorisation

This guideline has been authorised and issued by:

Authorised by	Dr John O'Hare - Deputy Director of Clinical Services
Authorisation date	03 11 2025
Published date	03 11 2025

19. Appendices

19.1 Appendix 1: ECT Consent



The voluntary adult patient with capacity- ECT may be given to voluntary adult patients with capacity if they have given their informed consent. Informed consent should be obtained for both the ECT and the anaesthetic.

Voluntary patients who have capacity to consent to treatment must provide valid informed consent to any treatment that is being proposed and they retain the right to withdraw consent at any time during treatment (MHA 2014 Part 5).

The voluntary adult patient with impaired capacity- An enduring guardian, guardian or “person responsible for the patient” (as defined in s. 110ZD of the Guardianship and Administration Act 1990) can provide informed consent to ECT for a voluntary adult patient who lacks capacity.

The involuntary adult patient - ECT may ONLY be given to an involuntary patient with the approval of the Mental Health Tribunal (MHT) unless under the emergency ECT provisions within the MHA 2014. The Mental Health Tribunal can only approve ECT being given at a mental health service approved under s. 544 to provide ECT. If the Mental Health Tribunal approves a course of ECT, the administration of anaesthetic is included as part of the ECT.

Valid informed consent is not necessary for **involuntary** or mentally impaired accused patients. The psychiatrist must still have regard to the patient’s wishes and, in ascertaining those wishes, must ensure that the patient is provided with the same explanation of the proposed treatment (MHA, 2014 Part 5)